CIE AS PHYSICS (9702) • QUESTION PAPER

9.1 Electric Current

Syllabus Section: 9. Electricity **Paper Type:** Structured Questions (Paper 2) **Total:** 7 Questions (62 Marks)

Question 1 Easy [11 marks]

Part (a) [2 marks]

Define the amp.

Part (b) [2 marks]

A charge of 60×10^{-6} C flows through a given section of a conductor in 140×10^{-3} s. Calculate the electric current, stating the final answer in mA.

Part (c) [4 marks]

Use words from the list below to complete the description of the movement of charge carriers in a conductor.

average current	**delocalised**	**electric force**
electric field	**randomly**	**drift**

The charge carriers in a metal conductor are ..... electrons.

Normally the electrons move ..... in all directions, but if a potential difference is applied between two points on the conductor, then an ..... is created.

This causes an ..... to act on the charge carriers, causing them to ..... along the conductor in a resultant direction.

Therefore we can say that a steady ..... flows through the conductor.

Part (d) **[3 marks]**

The conductor described in part (d) has a cross-sectional area of 1.1×10^{-7} m^2 and the electrons have an average drift speed of 18 mm s^{-1}.

Calculate the number density of electrons in this conductor.

Question 2 Easy [9 marks]

Part (a) [2 marks]

The Standard International System of Units (SI) is a system of measurement based on seven base units.

State the **symbol** and the **SI unit** for electric charge.

Part (b) [2 marks]

In a circuit, a current of 2.0 A flows through a resistor for 90 minutes.

Calculate the total charge passing through the resistor in this time.

Part (c) [2 marks]

Determine the number of electrons that pass a point in the resistor in part (b).

Part (d) [3 marks]

When a copper wire is exposed to a potential difference, a current is detected in it.

Explain, with reference to charge carriers and electric fields, why there is a current detected in the wire.

Question 3 Easy [11 marks]

Part (a) [3 marks]

Electric current is defined as the rate of flow of electric charge.

(i) State the particle responsible for carrying electric charge around a circuit

[1]

(ii) Calculate the total electric charge carried by ten million of these particles

[2]

Part (b) [3 marks]

A lamp forms part of a circuit. A current of 0.3 mA passes through the lamp for a period of 5 minutes.

Calculate the amount of charge which passes a point in the lamp over those 5 minutes.

Part (c) [2 marks]

A section of wire in the lamp in part (b) has a cross-sectional area of 4.5×10^{-5} m^2.

Using the current calculated in part (b), calculate the average drift velocity of the electrons when the number density of electrons in this section of wire is 3.6×10^{22} m^{-3}.

Part (d) [3 marks]

Fig. 1.1 shows a typical electric circuit involving a battery, lamp and two resistors.

There is a gap in the circuit between points **X** and **Y**.

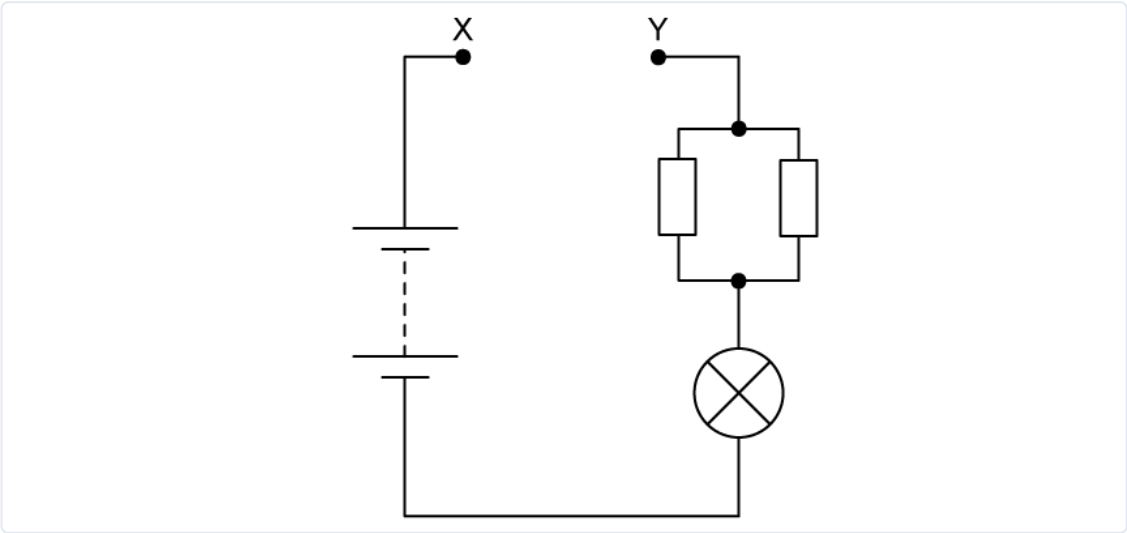


Fig. 1.1

Draw on Fig. 1.1:

(i) An **ammeter** between **X** and **Y** that would measure the current in the circuit

[2]

(ii) The direction of **conventional current** in the circuit

[1]

Question 4 Medium [11 marks]

Part (a) [4 marks]

The smallest conductor within a computer processing chip can be represented as a rectangular block that is one atom high, four atoms wide and twenty atoms long. One such block is shown in Fig. 1.1.

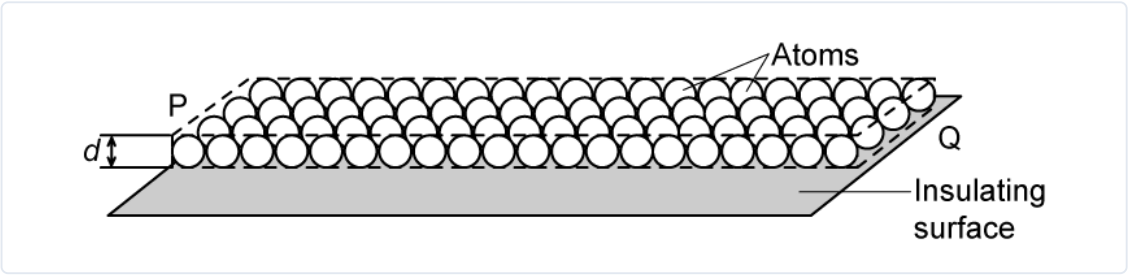


Fig. 1.1.

The block is made from phosphorus atoms of diameter $d = 3.8 \times 10^{-10}$ m. The atoms are deposited on an insulating surface. This ensures that the atoms touch each other.

The processing chip has a resistance, measured from **P** to **Q**, of 220 Ω.

(i) Define current.

[1]

(ii) Describe for this processing chip the properties which will affect the current.

[3]

Part (b) [1 mark]

Calculate the number density of free electrons within the block.

Assume that each phosphorus atom contributes one free electron.

Part (c) **[3 marks]**

Calculate the current between P and Q when the mean drift velocity of free electrons in the block is 1.9×10^{-5} m s^{-1}.

Part (d) **[3 marks]**

There are about 10^{9} of these tiny conductors in a single chip each carrying the current calculated in part (c).

Estimate the total power dissipated in these conductors in a single chip.

Question 5 Medium [6 marks]

Part (a) [1 mark]

A power source is connected across a component. The current in the component is 40 mA.

Calculate the total charge that flows past a point in the component in 3 minutes.

Part (b) [1 mark]

Calculate the number of electron charge carriers passing the same point in the conductor in this time.

Part (c) [4 marks]

8.6 J of energy are transferred to the conductor in this time.

(i) Calculate the potential difference across the component.

[2]

(ii) Calculate the resistance of the conductor.

[2]

Question 6 Medium [10 marks]

Part (a) [2 marks]

Nancy is testing a new type of light bulb. She records the amount of current and the relative brightness of the light coming from the bulb.

Sketch a possible circuit diagram to show this experimental setup.

Part (b) [3 marks]

Explain why the ammeter needs to be connected in series.

Part (c) [3 marks]

Explain the results Nancy should expect to obtain, assuming the resistance of the bulb, ammeter and circuit remain constant.

Part (d) [2 marks]

Nancy sets up her circuit and begins to conduct her experiment. Her first readings are negative values of current.

Identify the problem and suggest a solution.

Question 7 Medium [4 marks]

Part (a) [1 mark]

Define the *coulomb*.

Part (b) [1 mark]

An electric current is a flow of charge carriers.

In the following list, underline the possible charges for a charge carrier.

1.6×10^{-20} C 1.6×10^{-19} C 4.0×10^{-19} C 8.0×10^{-19} C

Part (c) [2 marks]

The diameter of a wire XY varies linearly with distance along the wire as shown in Fig. 6.1.

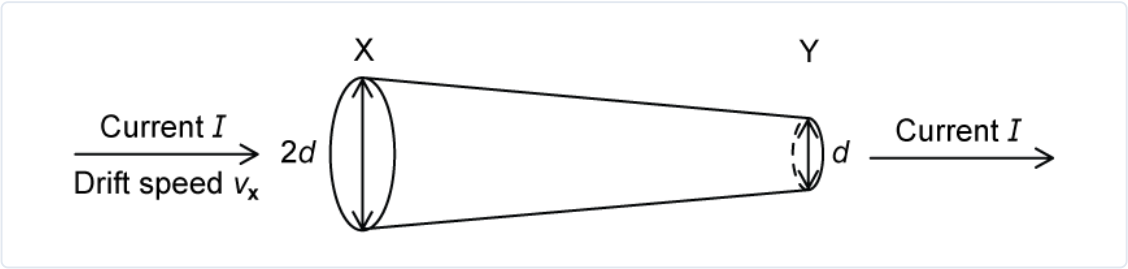


Fig. 6.1

There is a current I in the wire. At end X of the wire, the diameter is $2d$ and the average drift speed of the free electrons is v_X. At end Y of the wire, the diameter is d.

On Fig. 6.2, sketch a graph to show the variation of the average drift speed with position along the wire between X and Y.

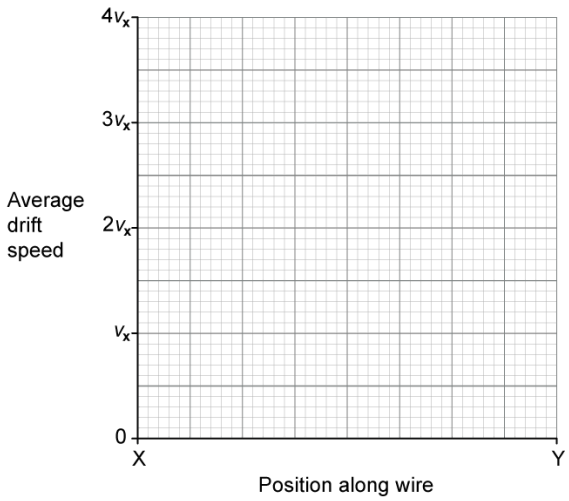


Fig. 6.2